GERARDO ANTONIO CORRAL RUIZ

AI/ML Applied Research & Advisory • Data Science • Master's candidate in Digital Innovation & Governance

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PROFESSIONAL SUMMARY

Master's candidate **(LM-91: Methods and Techniques for the Information Society)** at the **University of Bologna**, with a focus on **machine learning and artificial intelligence with a critical approach to governance, cyber risk management, and digital transformation**. I have developed **machine learning projects for industrial cybersecurity, particularly in IIoT/OT contexts**. I have also carried out university projects in **public health based on open data from sources such as ISTAT, Open-Meteo, and Kaggle**. I continually develop my **data science skills using Python and R**. Proficient in **Agile methodologies (Scrum)**, with experience using tools such as **Notion, Taiga, Essence Workbench, Slack, and Airtable**. As a **university lecturer**, I lead **contemporary philosophical debates with my students on technology and digital ethics**, linking these topics to the **interactions between technology, society, and governance.**

PROFESSIONAL EXPERIENCE

AI/ML Applied Research & Advisory @ Syntezia Sàrl (Switzerland) — 2026-Present

During my internship at **Syntezia**, I am conducting applied research on **Artificial Intelligence and Machine Learning for cybersecurity in IT and OT/IIoT environments**. I develop **Intrusion Detection System approaches based on Machine Learning and Deep Learning, using industrial telemetry and network traffic.**

This research is part of an initiative currently being prepared for an application to an international **European-level project**. In addition, I analyze where, when, and under which constraints the solution creates value across the **Protect/Detect/Respond functions of the Cybersecurity Framework 2.0**, while remaining aligned with a **governance, risk management, and operational priorities perspective.**

In parallel, I am carrying out an **ecosystem mapping activity involving EDIHs and potential partners**. In this context, I interact with different **stakeholders to understand their interests, identify possible synergies, and encourage their involvement in projects of shared interest and mutual value.**

University and High School Teacher (Mexico)

- ***Universidad Panamericana*** – Course on **Selected Topics in Contemporary Culture** (online lectures) — **2025-Present**
- ***Northridge School Mexico*** – Full-time **Spanish** and **Philosophy** courses. — **2023-2024**

Product Owner @ PCI Lab (Mexico) — 2020-2022

Development of a **serious game** with the Narrative Design Team **to promote emotional intelligence among 15–18-year-olds.** We adopted **Agile (Scrum) with weekly sprints, using Airtable for the backlog, roadmap, and Slack for real-time coordination.** The project received the **ILAN Innovation Award.**

EDUCATION

M.S. in Digital Innovation, Policies and Governance (LM-91) @ University of Bologna (Italy) — 2024-2026

I was **elected class representative** in my Master's program. To date, my **GPA is 29.43/30**. Below are a few key highlights from the **main projects I've developed**.

- **Machine Learning for Cardiovascular Disease Detection (*Laboratorio di Programmazione*, INF-05/A, 2024)** — **FINAL GRADE: 30/30**
  - End-to-end Data Science project on the Heart Disease Dataset (Kaggle): data cleaning, feature engineering, EDA, and training/evaluation of an SVM classifier to predict heart disease.
- **Moderate and Extreme Temperatures and Climate–Health Risk in Bologna: A Data-Driven Study of Climate Risk (*Metodi Informatici per la Trasformazione Digitale*, INF-05/A, 2025)** — **FINAL GRADE: 30/30 with Honours**
  - I developed a machine learning methodology to identify high-risk days for vulnerable groups, providing targeted support for public health prevention and policy decision-making.

- **The Digital Panopticon of Artificial Intelligence: between Safety and Freedom (*Governance e Diritto delle PPAA*, IUS/09, 2024)** **FINAL GRADE: 30/30 with Honours**
  - I analyzed the impact of AI on public safety in Europe, with a particular focus on the EU AI Act, the GDPR, and Regulation (EU) 2024/1689.
- **Cyber Threats in IIoT: Can ML Mitigate Vulnerabilities and Prevent Harm? (*Governance della Cybersecurity*, ING-INF/05, 2025)** **FINAL GRADE: 30/30 with Honours**
  - I wrote a paper on cyber risk and governance in IIoT/OT environments and developed a Python case study on ToN-IoT (Modbus telemetry), testing ML models (Decision Tree, Random Forest, SVM) for binary and multi-class attack classification, with hyperparameter tuning and evaluation (F1, FPR/FNR). The goal is to help SOC analysts reduce alert fatigue and enable managers to identify where security needs to be strengthened.

B.A. in Philosophy @ Universidad Panamericana (Mexico) 2018–2023

- Published an **article on generative AI** for El Universal [one of Mexico's leading newspapers] (2022): https://www.generacionuniversitaria.com.mx/tu-voz/dibujo-asistido-por-inteligencia-artificial-es-arte
- **President of the Student Council**, Faculty of Philosophy (Aug 2021 – Jun 2022)

B.Eng. in Digital Animation Engineering and Video Game Programming (not completed) @ UP, Mexico 2017–2019

CERTIFICATIONS, COURSES, CONFERENCES & OTHER EXPERIENCES

AI, DATA SCIENCE & BUSINESS ANALYTICS

Intelligenza artificiale e business intelligence: strumenti e algoretica – IFOA **2025**
EITC/AI/MLP Machine Learning with Python – European Information Technologies Certification Institute (EITCI) **2025**
Seminario LEEIA Lógica, Epistemología y Ética de la Inteligencia Artificial – Universidad Panamericana **2025**
Strumenti di data analysis e visualization – IFOA **2025**
EITC/CP/PPF Python Programming Fundamentals – European Information Technologies Certification Institute (EITCI) **2025**
Business Analytics with Excel: Elementary to Advanced – Johns Hopkins University **2023**
What is Data Science? – IBM **2023**
PredictionX: Omens, Oracles & Prophecies – HarvardX **2020**

MANAGEMENT & INNOVATION

Management: A Professional Future – IESE, University of Navarra, Barcelona **2023**
Introducción a la Dirección de Empresas (IDE) – IPADE Business School **2021**

DIGITAL ARTS, CREATIVITY & STORYTELLING

Story and Narrative Development for Video Games & Introduction to Game Design – California Institute of the Arts **2022**
Modern Arts & Ideas – MoMA Certificate **2021**
Studied Cinematography and Digital Post-Production – SAE Institute **2019-2020**

PUBLIC SPEAKING & RESEARCH

Conference Speaker: "Identidad líquida en tiempos de redes sociales" – Instituto Antonio Nariño, Colombia **2024**
Academic & Research Assistant – Dr. Héctor Zagal presso Universidad Panamericana **2018–2023**
Theatre UP – Universidad Panamericana **2017-2020**

HARD SKILLS

- **Agile Work:** Essence Workbench, Taiga, Notion, Slack, Airtable, Protégé, LaTeX
- **Data Analysis Tools:** Python (Pandas, Seaborn, Numpy, Sklearn, DoWhy), R, SQL, Git, JavaScript
- **Generative AI Tools:** ChatGPT, Gemini, NotebookLM, Sora, Midjourney, ComfyUI
- **Design and Video Editing Tools:** Adobe (Premiere, Photoshop, After Effects), Canva
- **Microsoft Office:** Word, PowerPoint, Excel

LANGUAGES

- **Native:** Spanish
- **Advanced:** English, Italian
- **Basic:** Japanese

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